

Discussion of “Tax Incentives and Venture Capital Risk-Taking” (Campello and Junqueira 2025)

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NBER Place-Based Policies and Entrepreneurship
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Roadmap of Talk

Context

Paper in One Slide

Economic Mechanisms

Empirics

Unanswered Question

Context: Tax Subsidies for Entrepreneurship

Two broad approaches:

- **Subsidizing startups:** R&D tax credit, accelerated depreciation, corporate income tax cut, payroll tax credit
- **Subsidizing investors of startups:** investor tax credit, capital gain tax credit/exemption (QSBS, Opportunity Zone)

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- **Subsidizing startups:** R&D tax credit, accelerated depreciation, corporate income tax cut, payroll tax credit
- **Subsidizing investors of startups:** investor tax credit, capital gain tax credit/exemption (QSBS, Opportunity Zone)
 - ▶ **Advantages**
 - ★ No need for government to pick startups → retain market incentives
 - ▶ **Disadvantages**
 - ★ Less direct targeting
 - ★ May lead to tax arbitrage if subsidy does not address frictions in investment

Context: Capital Gain Tax Credit vs Investor Tax Credit

- **Capital gain tax exemptions** have strong effects on
 - ▶ VC investments (Edwards and Todtenhaupt 2020)
 - ▶ Firm entry, innovation, employment (Chen and Farre-Mensa 2020, Arefeva et al. 2024)
 - ▶ Investor risk-taking (Campello & Junqueira 25)
- **Investor tax credits** have limited effect (Denes et al. 2023)
 - ▶ ↑ investment but no effect on growth or entry
 - ▶ Take-up by non-professional investors; crowd out non-subsidized financing;

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- **What explains the contrast?** (Appendix model of Denes et al. 2023)
 - ▶ Capital gain tax credit is state-contingent, while investor tax credit is flat
 - ▶ Both ↑ investment, but effect of investor tax credit ↓ with return skewness, while effect of capital gain tax credit ↑ with skewness

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This paper pushes the inquiry further by explicitly examining investor risk-taking

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- **Key findings:**
 - ▶ **QSBS matters:** bunching of holding period at 5 years; eligibility $\uparrow$ investments
 - ▶ **Eligible deals are riskier ex ante:** early-stage, more debt, out of state, new industry, first round, sole investor
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- **Methods:** model, bunching estimator, RDD, DID, DDD...
- **Contributions:**
 - ▶ Goes beyond the first-moment impact of QSBS to look at investor risk-taking
 - ▶ Novel finding that QSBS increases *pre-tax* returns, suggesting removal of some frictions

This Is an Important Paper on an Important Policy

SANTA CLARA BUSINESS LAW CHRONICLE

Incentivizing Innovation: How the One Big Beautiful Bill Act Supercharged QSBS



This Is an Important Paper on an Important Policy

carta



Contents: What is qualified small business stock (QSBS)?



What is qualified small business stock (QSBS)?

The qualified small business stock (QSBS) exclusion is a U.S. tax benefit that applies to eligible shareholders of a qualified small business (QSB). Because founding, investing in, and working for a startup can be riskier by nature, the QSBS exclusion helps encourage people to take that risk.

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Why Does QSBS Increase Investor Risk-Taking?

- Paper illustrates with a nice model that features investor heterogeneity
- Intuition: QSBS increases upside payoff without changing downside payoff $\Rightarrow \uparrow$ payoff convexity $\Rightarrow \uparrow$ risk-taking
- This effect is stronger if pre-tax payoff itself is more convex
 - ▶ Optionality: Convertible preferred $>$ common equity
(Paper models angels as common holder, but majority of them hold convertible notes)
 - ▶ Optionality: GP $>$ LP (carry is an option)

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(Paper models angels as common holder, but majority of them hold convertible notes)
 - ▶ Optionality: GP $>$ LP (carry is an option)
- But common equity holders also have convex payoffs from limited liability
 - ▶ **QSBS also incentivizes entrepreneurs and employees**
 - ▶ Chen and Farre-Mensa (2025): QSBS increases startups' use of equity compensation.

Alternative Mechanism 1: Incentives on Entrepreneurs & Employees

- QSBS increases the **supply** of risky startups by incentivizing entrepreneurs to enter and take risk (Porteba 1988)
- Table 5: deals shift riskier in eligible sectors post 2009
 - ▶ Could reflect change in deal supply rather than VCs' demand/preferences
 - ▶ Null results on angel & CVC deals do not fully rule out, as they have different targets and may face different deal supply

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- **Suggestions**
 1. Restrict to startups founded before 2009
 2. Use conditional logit to model deal selection within a defined choice set (McFadden 1976, Goyal et al. 2024)

Alternative Mechanism 2: QSBS as a Horizon Shock

- Key eligibility requirement: investments must be held for at least 5 years
 - ▶ Bunching result suggests a binding effect
- Longer investment horizon could explain
 1. A shift to pre-commercialization deals
 2. Greater dispersion in deal outcomes and a longer right tail
 3. Stronger response by funds with elite school endowment

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- Design:
 - ▶ Use initial round amount to proxy for size at next round.
 - ▶ Compare forward funding for initial rounds above vs below \$50mil

RDD Result on Forward Funding

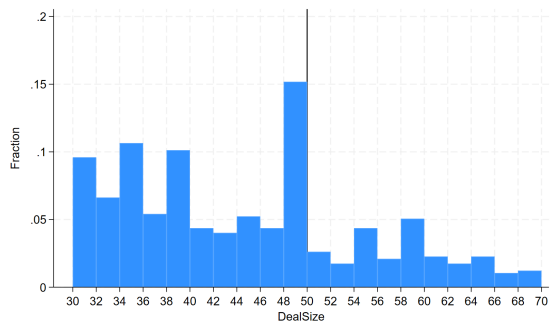
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Density of Initial Round Size in Eligible Industries Post 2009

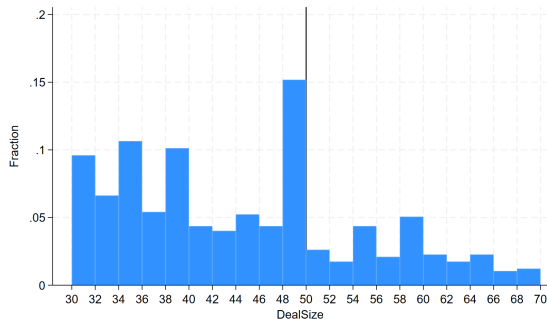
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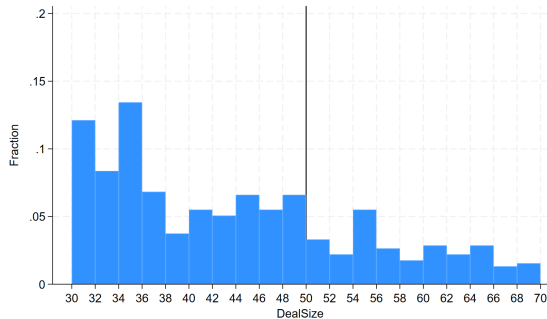
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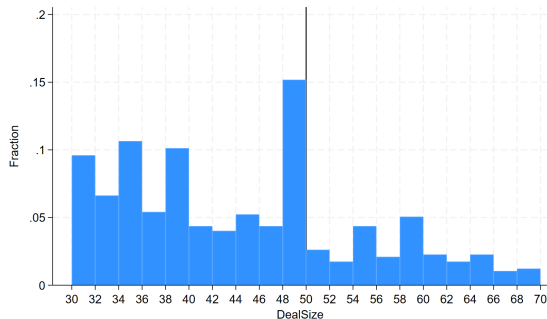
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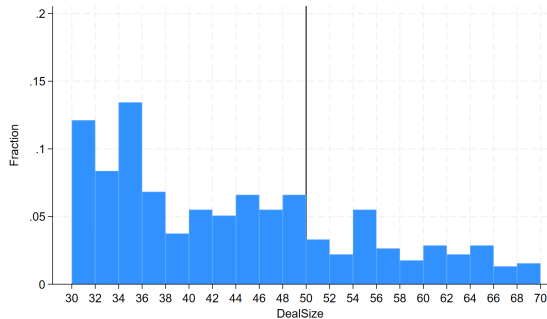
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Drop multiples of 10

- Suggest to drop RDD results. Investment effect is well documented anyways.

DDD Result on Deal Outcomes

	All Investors	
	All Firms (1)	Exited (2)
<i>Panel A: Dependent Variable = $1\{\text{Out of Business}\}$</i>		
$1\{\text{Eligible}\} \times 1\{\text{Holding} \geq 5\} \times 1\{\text{Post}\}$	0.011*** (5.04)	0.037*** (6.42)
$1\{\text{Eligible}\} \times 1\{\text{Holding} \geq 5\}$	0.001 (0.59)	-0.004 (-0.83)
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Observations	1,050,884	462,661
Adjusted R ²	0.11	0.11
Mean Dependent Variable	0.018	0.042
Year FE	✓	✓
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- Investor-firm FE absorbs deal selection, i.e. ex-ante risk-taking
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- Investor-firm FE absorbs deal selection, i.e. ex-ante risk-taking
- Remaining variation: ex-post risk-shifting?
 - ▶ Timing of success/failure within an investor-firm pair? But mechanical to bunching above 5 years
 - ▶ $\mathbb{1}(\text{Holding} \geq 5)$ is endogenous: more likely to hold beyond 5 years if expecting a big gain

DDD Result on Deal Outcomes

Suggestions:

1. Do this at the investor-firm level on a cross-section of deals. Remove investor-firm FE
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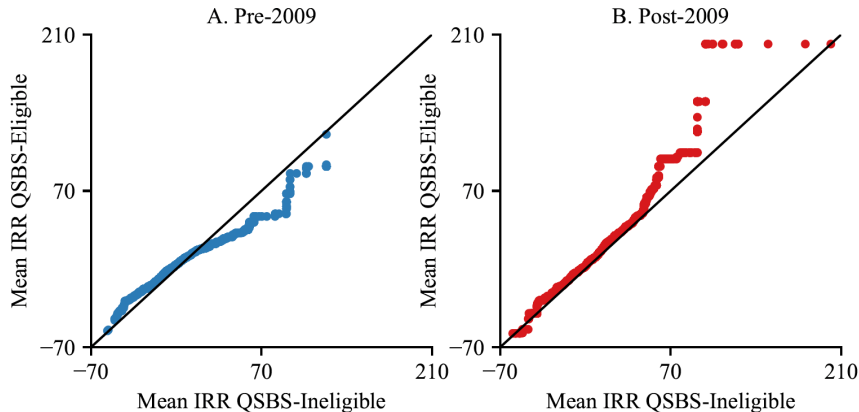
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 - ▶ $\mathbb{1}(\text{Holding} \geq 5)$ should be time-invariant, realized holding period
 - ▶ Outcomes are eventual success/failure
2. Focus on real outcomes (closure/IPO/M&A) rather than valuation
 - ▶ Concern: investors bid on eligible deals, pushing up valuation
 - ▶ Edwards & Todtenhaupt (2020): 2/3 of the tax benefits passed to startups through higher valuation.

Return Results

Figure 9: Portfolio Return Distributions: Enhanced QSBS Tax Benefits and VC Investment Outcomes



- Don't observe deal returns; so far proxied with fund returns
- Better approach: regress fund returns on %Eligible

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 - ▶ Return result suggests “money left on the table”
 - ▶ What prevented investors from taking more risk?

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 - ▶ What prevented investors from taking more risk?
- **Potential explanations:**
 - ▶ Innovation blindspots:
 - ★ Prior version shows that eligible deals are geographically more dispersed (**bring it back!**)
 - ★ **Suggest: test if eligible deals more likely to be in non-hub cities or outside GPs' network**
 - ▶ Risk-aversion/under-diversification of GPs or LPs
 - ▶ Impatience of GPs or LPs

Recap

- An impressive paper examining the effect of capital gain tax subsidies on VC risk-taking
- Punchline: **state-contingent tax incentives induce “healthy” risk-taking that improves returns and potentially capital allocation**
- Suggestions to tighten up empirics, rule out other explanations, and pinning down the frictions addressed by QSBS